All fifth-moment Berndt integrals: a corrected arithmeticity theorem MathDB 361365

Research note

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Task 1. Resolve the arithmeticity conjecture

For each integer $M \geq 1$, put

$$J_M = \int_0^\infty \frac{x^5\,dx}{(\cos x + \cosh x)^M}, \qquad G = \Gamma(1/4), \qquad u = \frac{G^4}{4\pi^3}.$$

We prove the even-power assertion of Xu–Zhao's Conjecture 5.9 [1, p. 27] for every $M = 2n$. The odd-power assertion as printed is false: its bulk upper limit $n-3$ omits an entire band. We prove the corrected assertion, with upper limit $n-2$, for every $M = 2n-1$, and give an exact counterexample to the printed assertion at $M = 7$. Thus the result includes a full all-power theorem, beyond identifying the indexing error. Empty sums below mean zero.

Theorem 1. *For every $M \geq 1$, the number J_M is a polynomial over $\mathbb{Q}$ in u and π. Every nonconstant monomial $u^j\pi^a$ occurring in this polynomial satisfies*

$$j > 0, \qquad j \equiv M \pmod 2, \qquad 1 \leq a \leq 6, \qquad j - a \leq M - 4. \tag{1}$$

In addition, $j = 1$ permits only $a = 1, 2$; $j = 2$ permits only $a = 1, 2, 3, 4$; and $(j, a) = (4, 5)$ never occurs. A rational constant is permitted only when M is even.

In the notation of the source, this proves, for every $n \geq 1$,

$$\begin{aligned} J_{2n-1} \in{}& \frac{G^4}{\pi}\mathbb{Q} + \frac{G^4}{\pi^2}\mathbb{Q} + \sum_{j=2}^{n-2}\sum_{i=1}^{6} \frac{G^{8j-4}}{\pi^{6j-10+i}}\mathbb{Q} \\ &+ \sum_{j=1}^{3}\sum_{i=1}^{7-2j} \frac{G^{8j+8n-20}}{\pi^{6j+6n-22+i}}\mathbb{Q}, \end{aligned} \tag{2}$$

$$\begin{aligned} J_{2n} \in{}& \mathbb{Q} + \sum_{i=1}^{4} \frac{G^8}{\pi^{i+1}}\mathbb{Q} + \sum_{\substack{1\leq i\leq 6 \\ i\neq 2}} \frac{G^{16}}{\pi^{i+5}}\mathbb{Q} \\ &+ \sum_{j=3}^{n-2}\sum_{i=1}^{6} \frac{G^{8j}}{\pi^{6j-7+i}}\mathbb{Q} + \sum_{j=1}^{3}\sum_{i=1}^{7-2j} \frac{G^{8j+8n-16}}{\pi^{6j+6n-19+i}}\mathbb{Q}. \end{aligned} \tag{3}$$

The bounds are inclusions, not assertions that every allowed coefficient is nonzero. This distinction also handles overlaps in the displayed spaces for small n.

1. Residues for an arbitrary denominator power

Let $a_0 = (1+i)/2$, $b_0 = (1-i)/2$, and $F(z) = z^5/(\cos z + \cosh z)^M$. The factorization

$$\cos z + \cosh z = 2\cosh(a_0 z)\cosh(b_0 z)$$

places the first-quadrant poles at $z_n = (1+i)t_n$, where $t_n = \pi(n+1/2)$ and $n \geq 0$. Integrate on the positive triangle with vertices $0, R, iR$, $R = 2\pi N$. On its oblique side $x + y = R$,

$$|\cosh(a_0 z)| = \cosh((x-y)/2) \geq 1, \qquad |\cosh(b_0 z)| \geq \sinh(R/2).$$

That side contributes $O_M(R^6 e^{-MR/2})$. Since $F(iy) = iF(y)$, the two axial sides together contribute $2\int_0^R F(x)\,dx$. Consequently

$$J_M = \pi i \sum_{n\geq 0} \operatorname{Res}_{z=z_n} F(z). \tag{4}$$

The integral is convergent; the residue series obtained next is absolutely convergent by exponential decay of hyperbolic secants and their derivatives.

Define rational coefficients

$$A_{M,h} = [v^{2h}] \left(\frac{v}{\sinh v}\right)^M.$$

Expanding $F(z_n + w)$ gives

$$\begin{aligned} J_M = (-1)^M 2^{3-M}\pi \sum_{h=0}^{\lfloor (M-1)/2 \rfloor} (-1)^h A_{M,h} \sum_{\ell=0}^{\min(5, M-1-2h)} \frac{\binom{5}{\ell}}{r!} \\ \cdot \sum_{n\geq 0} (-1)^{nM} t_n^{5-\ell} (\operatorname{sech}^M)^{(r)}(t_n), \qquad r = M-1-2h-\ell. \end{aligned} \tag{5}$$

Here and below a superscript in parentheses denotes ordinary differentiation of the whole expression in parentheses. To check the phase in (5), use

$$\cosh(a_0(z_n + w)) = i(-1)^n \sinh(a_0 w), \quad \cosh(b_0(z_n + w)) = \cosh(t_n + b_0 w),$$

and, when $2h + \ell + r = M - 1$,

$$2^{-M} i^{-M} a_0^{2h-M} (1+i)^{5-\ell} b_0^r = i(-1)^{M+1+h} 2^{3-M}.$$

Put $\delta = 0$ for odd M and $\delta = 1$ for even M. The identity

$$(\operatorname{sech}^m)'' = m^2 \operatorname{sech}^m - m(m+1)\operatorname{sech}^{m+2}$$

implies

$$\operatorname{sech}^M t = \sum_{k=0}^{(M-1-\delta)/2} C_{M,k} \left(\frac{d}{dt}\right)^{2k} \operatorname{sech}^{1+\delta} t, \quad \sum_k C_{M,k} T^{2k} = \frac{\prod_{\substack{1+\delta \leq j \leq M-2 \\ j \equiv 1+\delta \ (2)}} (j^2 - T^2)}{(M-1)!}. \tag{6}$$

An empty product is one.

2. Divisor interchange bounds the number of derivatives

For nonnegative integers P, Q of opposite parity set

$$\Phi_{P,Q}(q) = \sum_{k,v\geq 1} k^P v^Q q^{kv}, \quad D = q\frac{d}{dq}, \quad s = \min(P,Q), \quad \kappa = \frac{|P-Q|+1}{2}.$$

All the sums converge absolutely for $|q| < 1$. Interchanging k, v if necessary yields

$$\Phi_{P,Q} = D^s L_{2\kappa-1}, \qquad L_{2\kappa-1}(q) = \sum_{k\geq 1} \frac{k^{2\kappa-1}q^k}{1-q^k} = \frac{B_{2\kappa}}{4\kappa}(1 - E_{2\kappa}(q)). \tag{7}$$

The Eisenstein normalization is $q = e^{2\pi i\tau}$. For $\kappa \geq 2$, the classical modular-form structure theorem gives $E_{2\kappa} \in \mathbb{Q}[E_4, E_6]$, homogeneous of weight 2κ [2, Proposition 4]. Rationality follows from the rational Fourier coefficients and an invertible rational coefficient matrix for the monomial basis. For $\kappa = 1$ we use E_2 itself. Ramanujan's identities [3, Section 7] give

$$DE_2 = \frac{E_2^2 - E_4}{12}, \quad DE_4 = \frac{E_2E_4 - E_6}{3}, \quad DE_6 = \frac{E_2E_6 - E_4^2}{2}. \tag{8}$$

Thus the nonconstant part of (7) is a rational polynomial of weight $2w$, $w = \kappa + s = (P+Q+1)/2$. Its degree in E_2 is at most s when $\kappa \geq 2$, and at most $s+1$ when $\kappa = 1$.

For even M define

$$H_{P,Q} = \Phi_{P,Q}(q) - (2^P + 2^{Q+1})\Phi_{P,Q}(q^2) + 2^{P+Q+1}\Phi_{P,Q}(q^4), \qquad q = e^{-\pi}. \tag{9}$$

Selecting odd k and giving v the sign $(-1)^{v-1}$ shows that this is $\sum_{k \text{ odd}, v\geq 1} k^P v^Q (-1)^{v-1} q^{kv}$. For odd M define

$$T_{P,Q} = \frac{\Phi_{P,Q}(iq) - \Phi_{P,Q}(-iq)}{2i}, \qquad q = e^{-\pi/2}. \tag{10}$$

The coefficient multiplier is $\chi_4(kv) = \chi_4(k)\chi_4(v)$, where χ_4 is the nonprincipal character modulo four. Expanding $\operatorname{sech} t = 2\sum_{j\geq 0}(-1)^j e^{-(2j+1)t}$ and $\operatorname{sech}^2 t = 4\sum_{v\geq 1}(-1)^{v-1} v e^{-2vt}$ now gives, for $d \geq 0$,

$$\sum_{n\geq 0} t_n^P (\operatorname{sech}^2)^{(d)}(t_n) = \pi^P 2^{2+d-P}(-1)^d H_{P,d+1}, \tag{11}$$

$$\sum_{n\geq 0} (-1)^n t_n^P \operatorname{sech}^{(d)}(t_n) = \pi^P 2^{1-P}(-1)^d T_{P,d}. \tag{12}$$

All differentiations and interchanges here are justified by locally uniform absolute exponential convergence for $t_n \geq \pi/2$. In (5)–(6) these formulas use

$$P = 5 - \ell, \quad d = M - 1 - 2h - \ell + 2k, \quad Q = d + \delta. \tag{13}$$

In particular $0 \leq P \leq 5$, $Q \geq 0$, P, Q have opposite parity, and $Q \geq 1$ in the even case. Although d grows with M, the number s of modular derivatives in (7) is at most five.

3. Exact modular substitutions and parity

The classical lemniscatic value is $\theta_3(i)^4 = u$ [1, equation (2.3)]. The required substitutions are

τ	E_2	E_4	E_6
$i/2$	$6/\pi - 3u/2$	$33u^2/4$	$-189u^3/8$
i	$3/\pi$	$3u^2/4$	0
$2i$	$3/(2\pi) + 3u/8$	$33u^2/64$	$189u^3/512$
$(1+i)/4$	$12/\pi - 3iu$	$-33u^2$	$189iu^3$

(14)

For completeness, theta duplication [4], starting from $\theta_2(i)^4 = \theta_4(i)^4 = u/2$, gives at $2i$

$$(\theta_2^4, \theta_3^4, \theta_4^4) = u\left(\frac{3-2\sqrt{2}}{8}, \frac{3+2\sqrt{2}}{8}, \frac{1}{\sqrt{2}}\right).$$

Substitution in

$$E_4 = \tfrac{1}{2}(\theta_2^8 + \theta_3^8 + \theta_4^8), \quad E_6 = \tfrac{1}{2}(\theta_2^4 + \theta_3^4)(\theta_3^4 + \theta_4^4)(\theta_4^4 - \theta_2^4)$$

gives the last two modular weights at $2i$ and i. For weight two use

$$2E_2(2\tau) - E_2(\tau) = \tfrac{1}{2}(\theta_3(\tau)^4 + \theta_4(\tau)^4), \quad E_2(-1/\tau) = \tau^2 E_2(\tau) + \frac{6\tau}{\pi i}.$$

The latter gives $E_2(i) = 3/\pi$. The modular transformations of E_4, E_6 give the row $i/2$. Finally, $(1+i)/4 = -1/(2i-2)$ gives the last row by translation and inversion. The values at $(-1+i)/4 = -1/(2i+2)$ are obtained from this last row by replacing u with $-u$.

For a homogeneous polynomial of weight $2w$, its value in the first row of (14) is 4^w times its value in the third row with u replaced by $-u$. Because $2^{P+Q+1} = 4^w$, the first and third terms of (9) have an even sum in u; the middle term is also even. The constant part of L when $s = 0$ cancels: in the even case $Q \geq 1$, so $s = 0$ forces $P = 0$, and $1 - 2^P - 2^{Q+1} + 2^{P+Q+1} = 0$. Thus $H_{P,Q}$ contains only even powers of u. Equation (10) takes the odd part in u of the last row, divided by i, and therefore contains only odd powers of u, with rational coefficients in u, π^{-1}.

4. Proof of every support restriction

Consider a homogeneous modular monomial after substitution. Let t be the number of selected π^{-1} factors from the E_2 factors. Its power of u is $j = w - t$. Multiplication by the external factor π^{P+1} in (5), (11), and (12) gives the power $a = P + 1 - t$. Consequently

$$j = \frac{M+5+\delta}{2} - h - \ell + k - t, \quad a = 6 - \ell - t, \quad j - a = \frac{M-7+\delta}{2} - h + k \leq M - 4. \tag{15}$$

For $\kappa \geq 2$, $t \leq s \leq P$ implies $1 \leq a \leq 6$ and $j \geq \kappa$. For $\kappa = 1$, a positive $j = s + 1 - t$ likewise forces $t \leq s$. Parity was proved above.

The only possible $j = 0$ terms have $\kappa = 1$ and $t = s + 1$. They vanish in the odd case. In the even case their multiplier, apart from a common rational factor and π^{-s-1}, is

$$3^{s+1}\left(2^{s+1} - 2^P - 2^{Q+1} + 2^{P+Q-s}\right).$$

When $P = Q + 1$ this is zero. When $Q = P + 1$ the external factor π^{P+1} makes $a = 0$. This proves the assertion about constants.

If $j = 1$, only $\kappa = 1$ is possible. Then $t = s$, and $|P - Q| = 1$ gives $a = P + 1 - s \in \{1, 2\}$. If $j = 2$, then either $\kappa = 1$, $t = s - 1$, giving $a \in \{2, 3\}$, or $\kappa = 2$, $t = s$, giving $a \in \{1, 4\}$.

It remains to exclude $j = 4, a = 5$. The equalities imply

$$a = P - s + 5 - \kappa.$$

If $Q > P$, then $P = s$ and $a \le 4$. Otherwise $P - s = 2\kappa - 1$, so $a = 5$ forces $\kappa = 1$, $P = Q + 1$, and $t = s - 3 \ge 0$. Since $P \le 5$ and $Q \ge 1$, only $(P, Q) = (4, 3), (5, 4)$ remain. Direct use of (7), (8), and (14) gives the exact identities

$$H_{4,3} = \frac{3u^2}{16\pi^2}, \qquad H_{5,4} = \frac{5u^2}{8\pi^3}. \tag{16}$$

For a short way to reproduce this finite algebra, apply the derivation in (8) three or four times to $(1 - E_2)/24$, then take the three substitutions with multipliers respectively $(1, -32, 256)$ or $(1, -64, 1024)$. Neither result has a u^4 term. This establishes all assertions of Theorem 1.

To obtain (2), write $j = 2b - 1$. For $b \ge 2$, (1) permits all six a values when $b \le n - 2$; for $b = n - 1, n, n + 1$ it requires $a \ge 2, 4, 6$, respectively, and no larger b occurs. The $j = 1$ restriction supplies the first two terms. The same calculation with $j = 2b$ gives (3), using the exceptional restrictions at $j = 2, 4$. Any overlap for small n enlarges the displayed space and causes no problem. Finally use $u^j \pi^a = 4^{-j} G^{4j} \pi^{a-3j}$.

5. An exact counterexample to the printed odd assertion

The preceding formulas also give an effective exact evaluation for each M. For $M = 7$, they give

$$\begin{aligned} J_7 = {} & \frac{713\pi^6}{707788800} u^9 + u^7 \left(\frac{707\pi^6}{265420800} - \frac{7\pi^5}{442368} + \frac{103\pi^4}{5898240} \right) \\ & + u^5 \left(\frac{51331\pi^6}{2123366400} - \frac{839\pi^5}{10616832} + \frac{581\pi^4}{2949120} - \frac{91\pi^3}{442368} + \frac{11\pi^2}{196608} \right) \\ & + u^3 \left(\frac{75\pi^6}{524288} - \frac{1295\pi^5}{1179648} + \frac{114331\pi^4}{53084160} - \frac{1019\pi^3}{442368} + \frac{1253\pi^2}{884736} - \frac{7\pi}{24576} \right) \\ & + u \left(\frac{51331\pi^2}{8847360} - \frac{839\pi}{221184} \right). \end{aligned} \tag{17}$$

This calculation requires only $E_2, E_4, E_6, E_8 = E_4^2$ in (7). The accompanying rational-arithmetic certificate records all contributions indexed by (h, k) in (5)–(6); their sum is exactly (17). In particular the coefficient of $u^3\pi^6$ is $75/524288 \neq 0$.

For $n = 4$ the printed bulk sum $\sum_{j=2}^{n-3}$ is empty. Its remaining terms involve only u^1, u^5, u^7, u^9, with rational coefficients times powers of π. The established algebraic independence of π and $\Gamma(1/4)$ (Chudnovsky's theorem, also a consequence of Nesterenko's theorem; see [2, p. 85]) implies algebraic independence of π and u. Thus distinct monomials in u, π are linearly independent over $\mathbb{Q}$. No combination of the printed terms can equal (17). This proves the printed assertion false, while (2) supplies its proved correction. □

Task 2. Determine the exact nonzero support

The theorem bounds possible monomials. Determine which coefficients actually vanish for each denominator power, and whether all six coefficients in the added odd band are nonzero for every $n \ge 4$. Seek formulas that also determine their signs.

Task 3. Treat every numerator degree

Extend the divisor-interchange method to x^{4p+1} for arbitrary $p \geq 0$, without a lower bound on p in terms of M. Derive the complete support restrictions and explain the exceptional vanishing coefficients as uniform identities in modular forms.

Task 4. Resolve the first-moment conjecture

Prove the two all-power formulas in Xu–Zhao Conjecture 5.8 using the same method with numerator x. Determine whether the much smaller number of derivatives gives closed recurrences for the rational coefficients.

Task 5. Find an efficient recurrence in the power

Develop a recurrence for the fifth-moment polynomials in u, π as M increases. Determine whether it can preserve the proved support restrictions without first generating and cancelling the larger residue expressions.

References

[1] C. Xu and J. Zhao, *General Berndt-Type Integrals and Series Associated with Jacobi Elliptic Functions*, arXiv:2401.01385v2 (2024).

[2] D. Zagier, *Elliptic Modular Forms and Their Applications*, in *The 1-2-3 of Modular Forms*, Springer (2008), pp. 1–103, Proposition 4 and p. 85; author-hosted chapter.

[3] S. Ramanujan, *On certain arithmetical functions*, Trans. Cambridge Philos. Soc. **22** (1916), 159–184, Section 7; online transcription.

[4] NIST Digital Library of Mathematical Functions, Section 20.7, theta identities and transformations.

Verification and status. The proof for arbitrary M is the analytic and algebraic argument above, not an inference from finite computation. Run `verify_evaluation.py` for the exact finite certificate and separate numerical diagnostics. The full current source and dated bounded claim checks accompany this note. The authoring agent performed the proof audit; independent peer review has not occurred.